# RQ Answers

omitted

2024-04-05

```
glimpse(all_data)
```

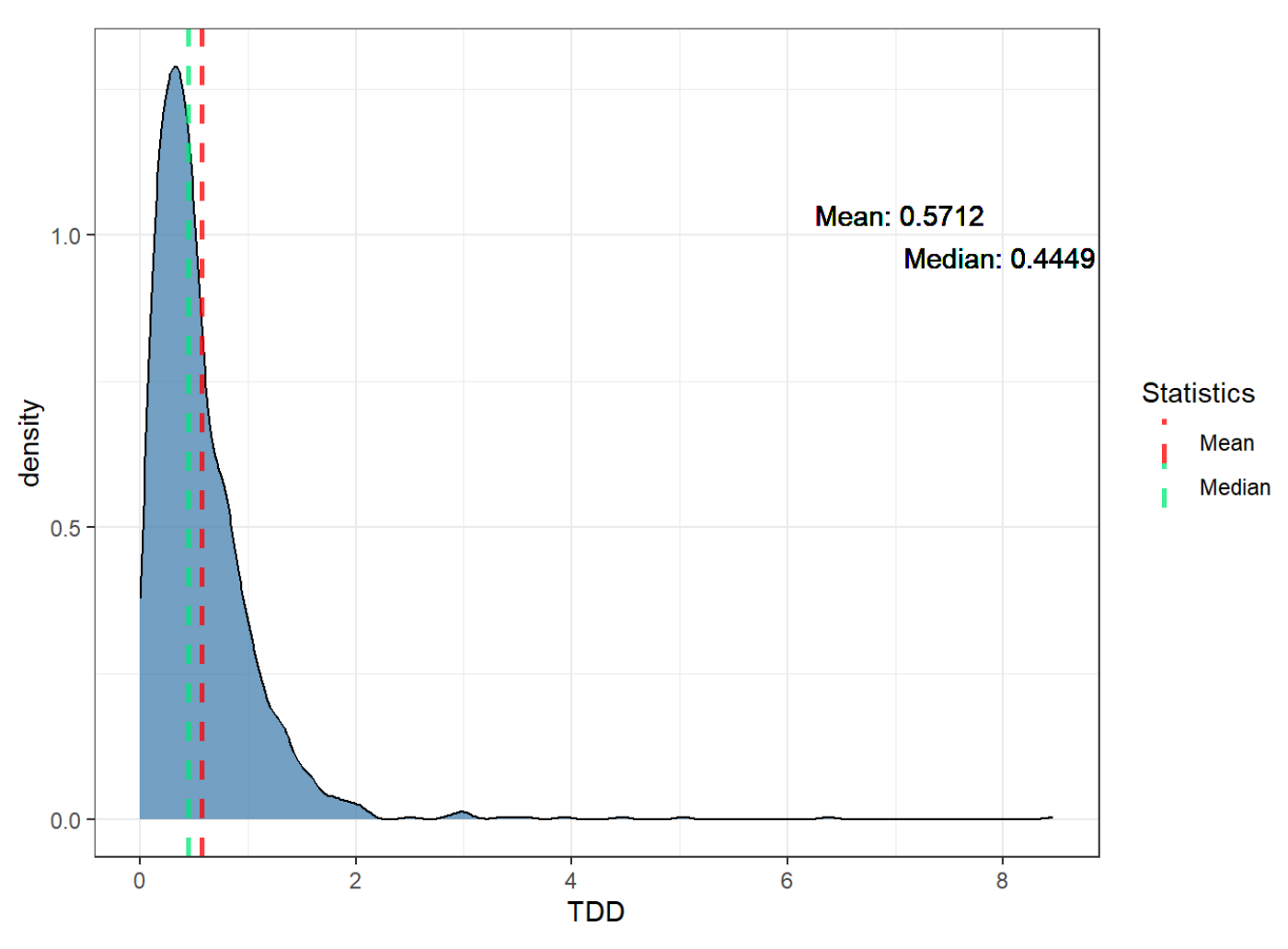
```
## Rows: 71,698
## Columns: 11
## $ repo               <chr> "accumulo", "accumulo", "accumulo", "accumulo", "a…
## $ pr_number          <int> 1302, 1302, 1302, 1302, 1302, 1302, 1302, 1302, 13…
## $ rule               <chr> "java:S5803", "java:S108", "java:S108", "java:S119…
## $ severity           <chr> "CRITICAL", "MAJOR", "MAJOR", "CRITICAL", "CRITICA…
## $ file               <chr> "server/tserver/src/test/java/org/apache/accumulo/…
## $ type               <chr> "CODE_SMELL", "CODE_SMELL", "CODE_SMELL", "CODE_SM…
## $ status             <chr> "OPEN", "CLOSED", "CLOSED", "OPEN", "OPEN", "OPEN"…
## $ debt               <int> 10, 5, 5, 10, 10, 10, 10, 10, 20, 10, 10, 5, 5, 20…
## $ ncloc_affected_file <int> 96, 96, 96, 96, 96, 121, 121, 121, 249, 249, 249, …
## $ complexity         <int> 7, 7, 7, 7, 7, 6, 6, 6, 37, 37, 37, 37, 37, 37, 37…
## $ origin             <chr> "NEW", "PRE-EXISTING", "PRE-EXISTING", "PRE-EXISTI…
```

# RQ1

## TDD (Technical Debt Density) by PR Distribution

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Mean:", round(mean(tdd), : All aesthetics have length 1, but the da
ta has 1152 rows.
## ℹ Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Median:", round(median(tdd), : All aesthetics have length 1, but th
e data has 1152 rows.
## ℹ Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```

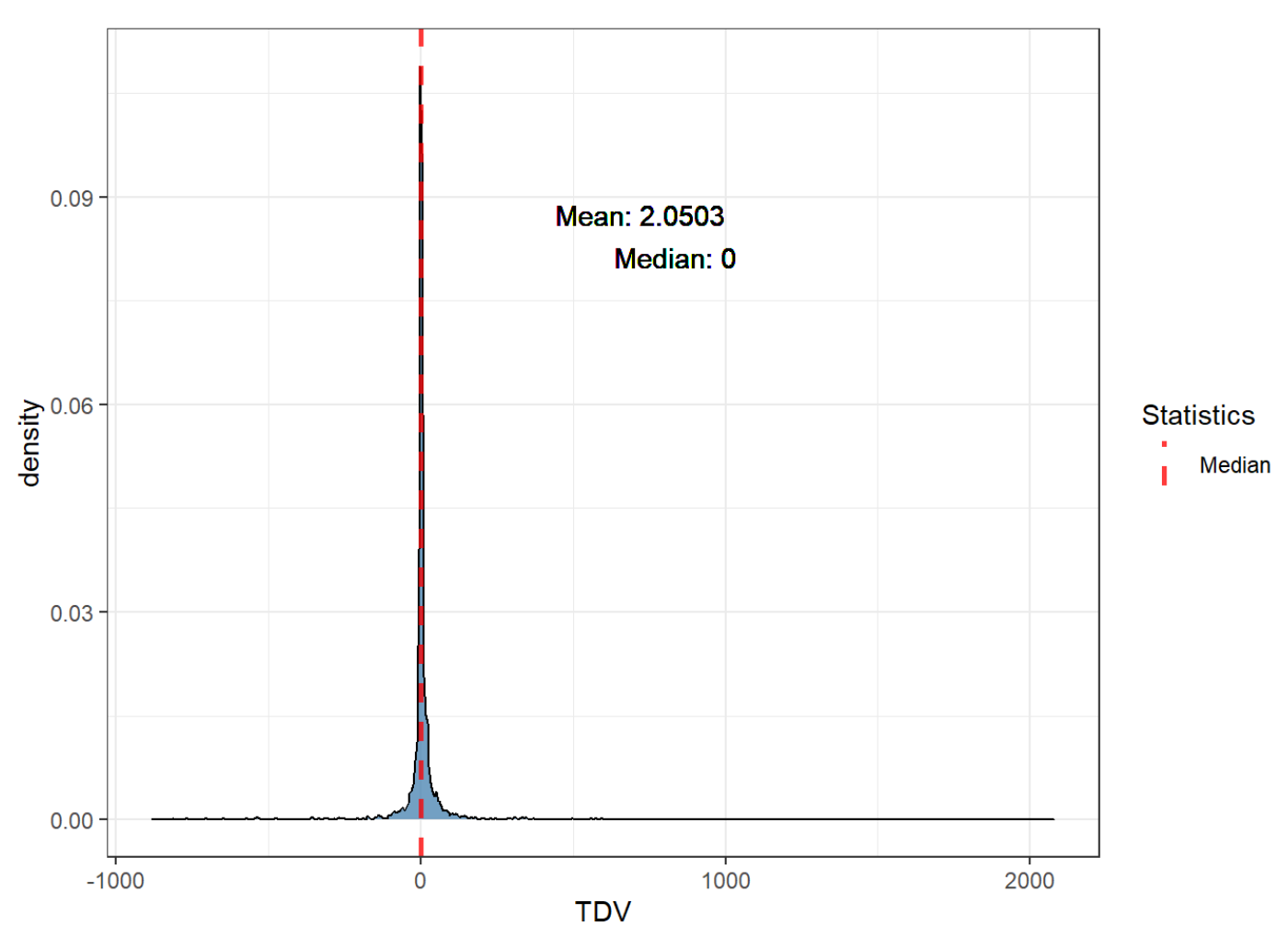


## TDV (Technical Debt Variation)

### TDV Distribution

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Mean:", round(mean(tdv), : All aesthetics have length 1, but the da
ta has 1152 rows.
## ℹ Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Median:", round(median(tdv), : All aesthetics have length 1, but th
e data has 1152 rows.
## ℹ Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```



### Percentages for TDV < 0 (TD decrease), TDV = 0 (TD unchanged) and TDV > 0 (TD increased) by repo

```
## # A tibble: 12 × 4
##    repo              tdv_lower_than_zero tdv_equal_to_zero tdv_greater_than_zero
##    <chr>                           <dbl>             <dbl>                 <dbl>
##  1 accumulo                        22.0               50.6                  27.3
##  2 cayenne                         11.1               40.7                  48.2
##  3 commons-collecti…               33.3               50                    16.7
##  4 commons-io                      28.8               50                    21.2
##  5 commons-lang                    21.8               63.6                  14.6
##  6 helix                           13.3               43.4                  43.4
##  7 httpcomponents-c…               16.7               43.8                  39.6
##  8 maven-surefire                  25                 65                    10
##  9 opennlp                         26.9               38.5                  34.6
## 10 struts                          26.7               36.7                  36.7
## 11 wicket                          12.5               50                    37.5
## 12 zookeeper                        9.09              54.6                  36.4
```

### Mean of percentages of TDV (mean of repo percentages)

```
## # A tibble: 1 × 3
##   mean_tdv_lower_than_zero mean_tdv_equal_to_zero mean_tdv_greater_than_zero
##                      <dbl>                  <dbl>                      <dbl>
## 1                     20.6                   48.9                       30.5
```

### Percentages of pre-existing TDV by repo

```
## # A tibble: 12 × 3
##    repo                    preexisting_tdv_equal_to_zero preexisting_tdv_greater…¹
##    <chr>                                           <dbl>                     <dbl>
##  1 accumulo                                         62.6                      37.4
##  2 cayenne                                          68                        32
##  3 commons-collections                              50                        50
##  4 commons-io                                       66.7                      33.3
##  5 commons-lang                                     69.1                      30.9
##  6 helix                                            69.8                      30.2
##  7 httpcomponents-client                            64.4                      35.6
##  8 maven-surefire                                   65                        35
##  9 opennlp                                          64.6                      35.4
## 10 struts                                           56.5                      43.5
## 11 wicket                                           61.5                      38.5
## 12 zookeeper                                        77.3                      22.7
## # ℹ abbreviated name: ¹preexisting_tdv_greater_than_zero
```

## Mean of percentages of pre-existing TDV (mean of repo percentages)

```
## # A tibble: 1 × 2
##   preexisting_tdv_equal_to_zero mean_preexisting_tdv_greater_than_zero
##                           <dbl>                                  <dbl>
## 1                          64.6                                   35.4
```

### Percentages for new TDV by repo

```
## # A tibble: 12 × 3
## # Groups:   repo [12]
##    repo                    new_tdv_equal_to_zero new_tdv_greater_than_zero
##    <chr>                                   <dbl>                     <dbl>
##  1 accumulo                                 83.0                     17.0
##  2 cayenne                                  93.3                      6.67
##  3 commons-collections                      80                       20
##  4 commons-io                               88.9                     11.1
##  5 commons-lang                             84.2                     15.8
##  6 helix                                    86.0                     14.0
##  7 httpcomponents-client                    95.6                      4.35
##  8 maven-surefire                          100                        0
##  9 opennlp                                  96.2                      3.85
## 10 struts                                   97.9                      2.08
## 11 wicket                                   90                       10
## 12 zookeeper                                80                       20
```

## Mean of percentages of new TDV (mean of repo percentages)

```
## # A tibble: 1 × 2
##   mean_new_tdv_equal_to_zero mean_new_tdv_greater_than_zero
##                        <dbl>                          <dbl>
## 1                       89.6                           10.4
```

# RQ2

As data is unbalanced across repositories, as shown in the issue characterization (../issues_characterization.Rmd), we will examine the top 10 fixed issues and the top 10 unfixed issues for each repository, and then aggregate them into a rank using the $PositionScore$ metric given by the following formula:

$$PositionScore_i = \sum_{j=1}^{k} Position_j$$

given a rule $i$, its $PositionScore_i$ will be the sum of its position across all $k$ repositiries.After calculating the metrics for all rules, we rank them in a TOP 10 of the most frequent rules (fixed or unfixed, depending on the context) across repositories. The lower the PositionScore, the more frequent the rule across repositories.

### TOP 10 fixed per repo ranked by count

All top 10 fixed rules for each repo.

```
## # A tibble: 10 × 13
## # Groups:   rank [10]
##     rank accumulo      cayenne `commons-collections` `commons-io` `commons-lang`
##    <int> <chr>         <chr>   <chr>                 <chr>        <chr>
##  1     1 java:S1192  … java:S… java:S112  -  10      java:S4719 … java:S4165  -…
##  2     2 java:S112  -… java:S… java:S1170  -  6      java:S1192 … java:S1192  -…
##  3     3 java:S1135  … java:S… java:S3008  -  5      java:S899  … java:S125  - …
##  4     4 java:S100  -… java:S… java:S1874  -  5      java:S4042 … java:S2293  -…
##  5     5 java:S2095  … java:S… java:S1134  -  4      java:S2129 … java:S3878  -…
##  6     6 java:S1155  … java:S… java:S125  -  4       java:S1874 … java:S1612  -…
##  7     7 java:S1659  … java:S… java:S1192  -  3      java:S2293 … java:S100  - …
##  8     8 java:S2259  … java:S… java:S1126  -  2      java:S1481 … java:S1611  -…
##  9     9 java:S1854  … java:S… java:S1172  -  1      java:S125  … java:S3984  -…
## 10    10 java:S3776  … <NA>    java:S2293  -  1      java:S1854 … java:S4968  -…
## # ℹ 7 more variables: helix <chr>, `httpcomponents-client` <chr>,
## #   `maven-surefire` <chr>, opennlp <chr>, struts <chr>, wicket <chr>,
## #   zookeeper <chr>
```

### TOP 10 fixed

TOP 10 fixed rules by PositionScore metric.

```
## # A tibble: 58 × 2
##    rule       position_score
##    <chr>               <dbl>
##  1 java:S1192             26
##  2 java:S112              70
##  3 java:S1874             74
##  4 java:S100              98
##  5 java:S125             106
##  6 java:S2293            112
##  7 java:S3740            115
##  8 java:S106             116
##  9 java:S1117            119
## 10 java:S1854            119
## # ℹ 48 more rows
```

### TOP 10 unfixed per repo ranked by count

All top 10 unfixed rules for each repo.

```
## # A tibble: 10 × 13
## # Groups:   rank [10]
##     rank accumulo      cayenne `commons-collections` `commons-io` `commons-lang`
##    <int> <chr>         <chr>   <chr>                 <chr>        <chr>
##  1     1 java:S1192  … java:S… java:S1192  -  295    java:S1192 … java:S1192  -…
##  2     2 java:S112  -… java:S… java:S125  -  207     java:S100  … java:S100  - …
##  3     3 java:S1186  … java:S… java:S108  -  173     java:S899  … java:S4165  -…
##  4     4 java:S1659  … java:S… java:S1117  -  109    java:S4042 … java:S125  - …
##  5     5 java:S2095  … java:S… java:S2293  -  78     java:S4719 … java:S1172  -…
##  6     6 java:S106  -… java:S… java:S2201  -  46     java:S2184 … java:S3878  -…
##  7     7 java:S100  -… java:S… java:S3878  -  45     java:S117  … java:S108  - …
##  8     8 java:S3252  … java:S… java:S1854  -  36     java:S108  … java:S3400  -…
##  9     9 java:S2184  … java:S… java:S112  -  35      java:S3878 … java:S1104  -…
## 10    10 java:S108  -… java:S… java:S1123  -  33     java:S1874 … java:S1186  -…
## # ℹ 7 more variables: helix <chr>, `httpcomponents-client` <chr>,
## #   `maven-surefire` <chr>, opennlp <chr>, struts <chr>, wicket <chr>,
## #   zookeeper <chr>
```

### TOP 10 unfixed

TOP 10 unfixed rules by PositionScore metric.

```
## # A tibble: 51 × 2
##    rule       position_score
##    <chr>               <dbl>
##  1 java:S1192             14
##  2 java:S100              86
##  3 java:S106              94
##  4 java:S112             103
##  5 java:S1186            103
##  6 java:S1874            106
##  7 java:S125             109
##  8 java:S2095            110
##  9 java:S108             112
## 10 java:S117             115
## # ℹ 41 more rows
```

# Outliers

## Higher TDD

```
## # A tibble: 1,152 × 5
##    repo           pr_number total_debt total_ncloc   tdd
##    <chr>              <int>      <int>       <int> <dbl>
##  1 wicket               709       2218         262  8.47
##  2 wicket               708       1161         182  6.38
##  3 struts               807        196          39  5.03
##  4 opennlp              563       1035         232  4.46
##  5 maven-surefire       596       1246         317  3.93
##  6 opennlp              569       1161         325  3.57
##  7 opennlp              585       1296         386  3.36
##  8 commons-lang        1051        978         324  3.02
##  9 helix               2439        102          34  3
## 10 struts               594        300         101  2.97
## # ℹ 1,142 more rows
```

## Higher TDV

```
## # A tibble: 1,152 × 3
##    pr_number repo                    tdv
##        <int> <chr>                 <int>
##  1      2503 accumulo               2078
##  2      1891 accumulo                643
##  3       483 httpcomponents-client   593
##  4      1605 accumulo                574
##  5      2335 accumulo                557
##  6       559 struts                  496
##  7      2680 helix                   368
##  8      2289 accumulo                345
##  9      1452 helix                   343
## 10       427 struts                  333
## # ℹ 1,142 more rows
```

## Lower TDV

```
## # A tibble: 1,152 × 3
##    pr_number repo                    tdv
##        <int> <chr>                 <int>
##  1       484 httpcomponents-client  -882
##  2       431 opennlp                -813
##  3      3604 accumulo               -768
##  4      1565 accumulo               -704
##  5       799 struts                 -648
##  6      3117 accumulo               -571
##  7       670 struts                 -542
##  8       826 struts                 -536
##  9       559 opennlp                -528
## 10      1584 helix                  -478
## # ℹ 1,142 more rows
```

# Extra

## Number of PRs with pre-existing TD

```
##      n
## 1 1094
```

## Number of java:S1192 issues that affect test code

```
##       n
## 1 18504
```

## Number of java:S1192 issues that affect not test code

```
##    n
## 1 84
```

## Summary NCLOC

```
##      ncloc
##  Min.   :    3
##  1st Qu.:  180
##  Median :  414
##  Mean   : 1107
##  3rd Qu.: 1024
##  Max.   :61510
```